

Computational Thinking and the Foundation of Artificial Intelligence

Concepts, brief history of AI and building intelligent systems

Dr. Elena Fedorchenko, Associate Professor

CUMT

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Task 6: Integrated scenario

Propose your own case and apply computational thinking to it.

Task 6: some common mistakes

Decomposition of the system vs decomposition of the problem/task.

Case: Fan monitoring

System decomposition

What parts does the system contain?

fan, temperature sensor, vibration sensor, motor, controller

nouns/objects

Problem decomposition

What subtasks must be solved?

collect measurements -> monitor fan condition -> detect abnormal behavior -> generate warning -> inform operator

actions/tasks

Task 6: some common mistakes

Decomposition vs Algorithms

Case: Monitor the condition of a mine ventilation fan.

Decomposition

What smaller tasks must be solved?

Answers: What needs to be done?

May not have a strict order.

Algorithm

In what order should the steps be performed to solve one task?

Answers: How should it be done?

Usually has an order.

Task 6: some common mistakes

Decomposition vs Algorithms

Case: Monitor the condition of a mine ventilation fan.

Decomposition

- Acquire sensor data.
- Validate and prepare the data.
- **Assess the fan condition.**
- Communicate the result.

Algorithm

Input: validated temperature and vibration measurements.

- 1 Compare temperature with the normal range.
- 2 Compare vibration with the normal range.
- 3 Evaluate the results together.
- 4 Classify fan condition as Normal/Warning/Critical.
- 5 Return the condition.

Sources

- *Michael Negnevitsky. Artificial Intelligence: A Guide to Intelligent Systems (3d Edition), 2011*
- *Course on Foundations of Artificial Intelligence by Dr. Ryan Urbanowicz, University of Pennsylvania (2020)*
(<https://www.youtube.com/watch?v=NVYSnFZwMKk&list=PLafPhSv10SDew4qhV9YeNFwf-1eUONHng>)
- Other

Course structure

- Introduction
- Computational thinking and intelligent systems
 - Computational thinking
 - Concepts, brief history of AI and building intelligent systems
 - Data, search and problem solving in intelligent engineering systems
- Knowledge representation and reasoning
 - Propositional logic for engineering decisions
 - First-order logic and knowledge representation
 - Rule-based expert systems
 - Frame-based expert systems, semantic networks and knowledge graphs
 - Ontologies and structured knowledge
- Search, uncertainty and intelligent decision-making
 - Information retrieval, semantic search and generative AI
 - Uncertainty and probabilistic reasoning
 - Fuzzy logic and fuzzy sets
 - Fuzzy inference and fuzzy expert systems
 - Optimization and evolutionary computation

From computational thinking to artificial intelligence

Computational thinking

- Helps us formulate a problem clearly.

Artificial intelligence

- Provides methods that can help solve some of these problems automatically.

- 1 The meaning of intelligence
- 2 Brief history of artificial intelligence
- 3 Different ways to build intelligent systems
- 4 AI applications in Mining
- 5 Building an intelligent system
- 6 Conclusion

Meaning of intelligence

Intelligence

- **Intelligence** is the ability to think and understand instead of doing things by instinct or automatically

Essential English Dictionary, Collins, London, 1990

Thinking

Thinking

- **Thinking** is the activity of using your brain to consider a problem or to create an idea

Essential English Dictionary, Collins, London, 1990

Intelligence

- **Intelligence** is ability to learn and understand, to solve problems and to make decisions

Meaning of artificial intelligence

Artificial intelligence

- **Artificial intelligence** is the field concerned with building systems capable of performing tasks that normally require aspects of human intelligence.

The goal of artificial intelligence

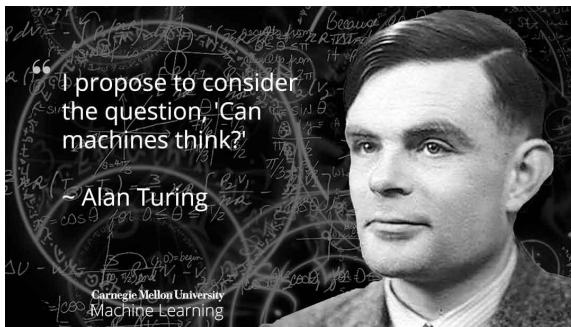
The goal of AI

- To make machines do things that would require intelligence if done by humans

Boden, 1977

Can machines think?

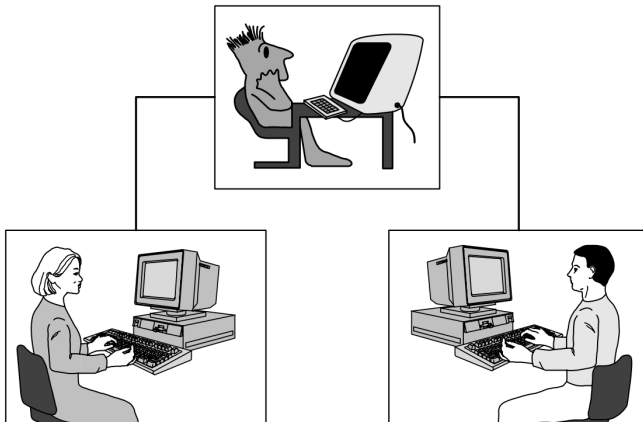
1950s: A. Turing has written one of the earliest and most significant papers on machine intelligence, 'Computing machinery and intelligence' .



<https://medium.com/@jetnew/a-summary-of-alan-m-turings-computing-machinery-and-intelligence-fd714d187c0b>

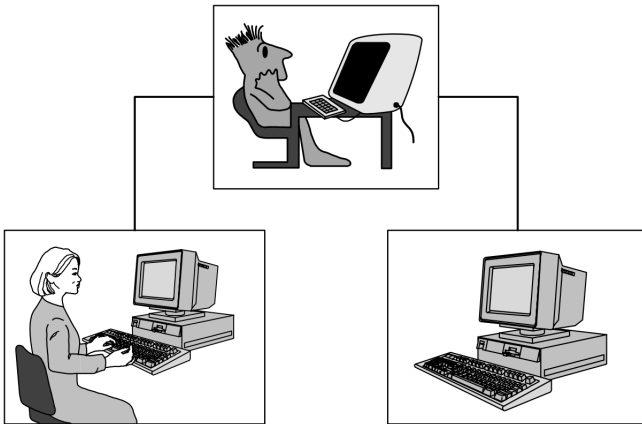
Turing imitation game

- Can machines pass a behaviour test for intelligence?



Michael Negnevitsky. *Artificial Intelligence: A Guide to Intelligent Systems (2d Edition)*, 2005

Turing imitation game: second phase



Michael Negnevitsky. Artificial Intelligence: A Guide to Intelligent Systems (2d Edition), 2005

Turing imitation game

I believe that in about fifty years' time it will be possible to programme computers, with a storage capacity of about 10^9 , to make them play the imitation game so well that an average interrogator will not have more than 70 per cent chance of making the right identification after five minutes of questioning.

—Turing, 1950

Turing imitation game: examples

- <https://humanornot.so/>
- <https://www.humanornot.ai/>
- <https://gandalf.lakera.ai/intro>

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Brief history of artificial intelligence

- 1943–1956: the birth of artificial intelligence (the 'Dark Ages')
- 1956–late 1960s: the rise of artificial intelligence (the era of great expectations)
- late 1960s–early 1970s: unfulfilled promises (or the impact of reality)
- early 1970s–mid-1980s: the technology of expert systems (or the key to success)
- mid-1980s–onwards: the rebirth of neural networks (how to make a machine learn)
Parallel approaches: fuzzy systems, evolutionary computation, probabilistic methods
- 2010s–Deep Learning
- 2020s: Foundation models and Generative AI

The birth of artificial intelligence

- **1943:** McCulloch & Walter Pitts – a model of artificial neural networks – each neuron was postulated as being in binary state, in either on or off condition (*McCulloch W.S., Pitts W. A logical calculus of the ideas immanent in nervous activity. Bulletin of Mathematical Biophysics 5, 115–133. 1943*).

The birth of artificial intelligence

- **1950:** Alan Turing has written one of the earliest and most significant papers on machine intelligence, ‘Computing machinery and intelligence’, where he considers the question ‘Can machines think?’.

VOL. LIX. No. 236.]

[October, 1950

MIND

A QUARTERLY REVIEW

OF

PSYCHOLOGY AND PHILOSOPHY



I.—COMPUTING MACHINERY AND INTELLIGENCE

By A. M. TURING

1. *The Imitation Game.*

I PROPOSE to consider the question, ‘Can machines think?’ This should begin with definitions of the meaning of the terms ‘machine’ and ‘think’. The definitions might be framed so as to reflect so far as possible the normal use of the words, but this attitude is dangerous. If the meaning of the words ‘machine’ and ‘think’ are to be found by examining how they are commonly used it is difficult to escape the conclusion that the meaning

<https://x.com/docmilanfar/status/1801315120083308634>



The birth of artificial intelligence

- **1950:** Claude Shannon – a typical chess game involved about 10^{120} possible moves (*C. Shannon. Programming a Computer for Playing Chess. Philosophical Magazine, Ser.7, 41, no. 314*). Shannon demonstrated the need to use heuristics in the search.

Heuristics: Computer chess

- Chess became a classic testbed for AI problem solving:
 - It is a finite game.
 - A brute-force algorithm going through all possible plays of chess is practically infeasible $\Rightarrow$ need for heuristics.
- 1958: Simon & Newell predicted that within 10 years the world chess champion would be a computer.
- 1997: Deep Blue (a chess-playing machine, IBM) defeated G. Kasparov.



Deep Blue – Kasparov Match. 1996

Today's chess-playing machines (AI systems) apply a heuristic approach to eliminate unlikely moves and focus on those with a high probability of defeating their opponent.

Hjeij M., Vilks A. A brief history of heuristics: how did research on heuristics evolve? Humanit Soc Sci Commun 10: 64. 2023

The birth of artificial intelligence

Intelligent problem solving often requires searching among many alternatives efficiently.

The birth of artificial intelligence

- **1956:** Minsky, Shannon & John McCarthy – workshop on machine intelligence, artificial neural nets and automata theory at Dartmouth College – birth of a new science – artificial intelligence.

The birth of artificial intelligence

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The rise of artificial intelligence

- McCarthy defined the high-level language LISP.
- **1958:** McCarthy – the Advice Taker – a program for searching the solutions to general problems of the world – the 1st complete KBS (*McCarthy J. Programs with common sense, Proc. of the Symposium on Mechanisation of Thought Processes, vol. 1, London, pp. 77–84, 1958*).
- **1975:** Minsky – the theory of frames (*Minsky M.L. A framework for representing knowledge, The Psychology of Computer Vision, P. Winston, ed., McGraw-Hill, New York, pp. 211–277, 1975*).

The rise of artificial intelligence

- **1961, 1972:** Allen Newell & Herbert Simon – the General Problem Solver – a general purpose program to simulate human problem-solving methods (Newell A., Simon H.A. *GPS, a program that simulates human thought, Lernende Automatten*, H. Billing, ed., R. Oldenbourg, Munich, pp. 109–124. 1961, Newell A., Simon H.A. *Human Problem Solving*. Prentice Hall, Englewood Cliffs, NJ. 1972).

The rise of artificial intelligence

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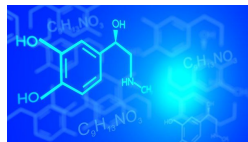
Unfulfilled promises

The main difficulties for AI in the late 1960s:

- Complexity of algorithms
- Too broad and too difficult problems.
- **1971:** Sir James Lighthill review the current state of AI – no results from AI research (*Lighthill J. Artificial intelligence: a general survey, Artificial Intelligence: A Paper Symposium. J. Lighthill, N.S. Sutherland, R.M. Needham, H.C. Longuest-Higgins, D. Michie, eds, Science Research Council of Great Britain, London, 1973.*).

The technology of expert systems

- **1965:** DENDRAL program (determine the molecular structure of Martian soil based on the mass spectral data provided by a mass spectrometer) – the 1st successful KBS: Edward Feigenbaum, Bruce Buchanan, Joshua Lederberg.
- shift from general purpose, weak methods to domain-specific, knowledge intensive techniques;
- using heuristics in the form of high-quality specific rules, the DENDRAL team proved that computers could equal an expert in narrow, defined, problem areas;
- originated the fundamental idea of the new methodology of expert systems – knowledge engineering (capturing, analysing and expressing in rules an expert's 'know-how').



<https://wrytin.com/ishani1/dendral-and-mycin-jwp32dfw>

The technology of expert systems

- **1972:** MYCIN project – provided a doctor with therapeutic advice: Feigenbaum, Edward Shortliffe and others.

- could perform at a level equivalent to human experts in the field;
- 450 independent rules of IF-THEN form;
- the knowledge was clearly separated from the reasoning mechanism;
- reasoning in the face of uncertainty.



<https://archive.org/details/rulebasedexperts00buch>

- **1974–1983:** PROSPECTOR – expert system for mineral exploration: Stanford Research Institute.

- rules and a semantic network to represent knowledge;
- over a 1000 rules to represent extensive domain knowledge;
- can explain the steps used to reach the conclusion.



<https://www.sri.com/hoi/prospector-computer-based-expert-system/>

The technology of expert systems: the difficulties

- Expert systems are restricted to a very narrow domain of expertise.
- Because of the narrow domain, expert systems are not so robust and flexible.
- Expert systems have limited explanation capabilities.
- Expert systems are difficult to verify and validate.
- Expert systems have little or no ability to learn from their experience.

The rebirth of neural networks

The rise of Neural Networks:

- **1943:** McCulloch and Pitts proposed an early artificial neuron model.
- **1950s–1960s:** perceptron and early neural network models.
- **1980s: renewed interest due to the multilayer networks and backpropagation.**
- **2010s:** deep learning achieved major progress in vision, speech and other data-driven tasks.

Neural Networks shifted AI from explicitly programmed rules toward learning patterns from data.

Parallel approaches in AI

Several important AI approaches developed in parallel with neural networks.

- **Fuzzy systems:** reasoning with vague and linguistic concepts such as *low*, *medium*, *high*.
- **Evolutionary computation:** optimization inspired by natural evolution: selection. mutation, crossover.
- **Probabilistic methods:** reasoning and decision-making under uncertainty.

Machine Learning and Deep Learning

- **1990s–2000s:** statistical machine learning became increasingly important;
- **2010s:** deep learning achieved major progress in image recognition, speech and other complex data;
- increasing computing power and larger datasets accelerated this development.

AI systems increasingly learned patterns from data instead of relying only on explicitly programmed rules.

Foundation models and generative AI

- **2017:** Transformer architecture (uses attention to identify relationships between elements of input data);
- large language models and foundation models (large AI model trained on broad data and designed to support many different tasks);
- generation of text, images and other content;
- multimodal AI;
- AI assistance, RAG (retrieval-augmented generation, combines information retrieval with a generative AI model to produce answers based on external sources) and agents.

Modern AI can not only classify or predict, but also generate, retrieve, reason and interact with users.

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Different ways to build intelligent systems

Approach

Rule-based / knowledge-based systems

Fuzzy systems

Machine learning / neural networks

Evolutionary computation

Probabilistic methods

Generative AI / foundation models

Main idea

Use explicit human knowledge

Reason with imprecise concepts

Learn patterns from data

Search for good solutions through optimization

Reason under uncertainty

Generate and transform complex content, interact with users

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Where can AI help in Mining?

Safety monitoring:

- methane and gas hazard detection;
- worker safety and PPE monitoring.

Mineral exploration:

- analysis of geological and geophysical data;
- support for mineral prospecting.

Production and process optimization:

- route and schedule optimization;

Equipment condition monitoring:

- fault detection;
- predictive maintenance.

Autonomous and intelligent equipment:

- autonomous haulage and drilling;
- obstacle detection and navigation.

Decision support:

- combining sensor data, expert knowledge and historical information;

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Davis's law

For every tool there is a task perfectly suited to it.

Davis R. & King J. An overview of production systems, Machine Intelligence, 8, 300–322, 1977

Building an intelligent system: basic phases

- 1 Problem assessment
- 2 Data and knowledge acquisition
- 3 Select an appropriate AI approach
- 4 Development of a prototype system
- 5 Development of a complete system
- 6 Evaluation and revision of the system
- 7 Integration and maintenance of the system

Building an intelligent system: basic phases

Phase 1: Problem assessment

- Determine the problem's characteristics
- Identify the main participants in the project
- Specify the project's objectives
- Determine the resources needed for building the system

Phase 1: Problem assessment

Problem type	Description	Example
Diagnosis	Inferring malfunctions of an object from its behaviour and recommending solutions	Identify a possible equipment fault
Selection	Recommending the best option from a list of possible alternatives	
Prediction	Predicting the future behaviour of an object from its behaviour in the past	Predict equipment condition or methane trend
Classification	Assigning an object to one of the defined classes	Classify operating condition
Clustering	Dividing a heterogeneous group of objects into homogeneous subgroups	Group similar operating patterns
Optimisation	Improving the quality of solutions until an optimal one is found	optimize route / ventilation / scheduling
Control	Governing the behaviour of an object to meet specified requirements in real-time	adjust an engineering process
Detection	Detect abnormal patterns	Detect abnormal or hazardous condition

Building an intelligent system: basic phases

Phase 2: Data and knowledge acquisition

- Collect and analyse data and knowledge
- Make key concepts of the system design more explicit

Phase 2: Data and knowledge acquisition

An intelligent system may use:

- sensor data,
- historical records,
- images,
- text/documents,
- expert knowledge.

Data quality matters: data may be noisy, incomplete or inconsistent.

Building an intelligent system: basic phases

Phase 3: Select an appropriate AI approach

Consider:

- Problem type: diagnosis, selection, prediction, classification, clustering, optimization, control, detection.
- Available data and knowledge: sensor data, historical records, images, text/documents, expert knowledge.
- Need for explainability: must the system explain the decision?
- Amount and quality of data: is enough reliable training data available?
- uncertainty and variability: are the inputs noisy, incomplete or imprecise?
- Operational constraints: real-time response, computing resources, safety requirements.
- Need for adaptation or learning: should the system improve from new data?

Phase 3: Select an appropriate AI approach

What information is available?

Data type

Expert rules / domain knowledge

Numerical or sensor data

images / video

text / documents

imprecise linguistic knowledge

uncertain information

Approach

rule-based methods

machine learning

computer vision

NLP, LLMs, RAG

fuzzy logic

probabilistic methods

Phase 3: Select an appropriate AI approach

Example: Ventilation Fan Monitoring

Data type

Clear expert rules available
Large historical sensor dataset
vague concepts such as *high vibration* or *slightly elevated temperature*
uncertain sensor measurements

Approach

Rule-based system
machine learning
fuzzy system
probabilistic methods

There is no universally best AI method. The choice depends on the problem, data and operational requirements.

Building an intelligent system: basic phases

Phase 3: Development of a prototype system

- Choose a tool for building an intelligent system
- Transform data and represent knowledge
- Design and implement a prototype system
- Test the prototype with test cases

Phase 4: Development of a prototype system

Prototype

A small version of the final system

Test case

A problem successfully solved in the past for which input data and an output solution are known

Building an intelligent system: basic phases

Phase 5: Development of a complete system

- Prepare a detailed design for a full-scale system
- Collect additional data and knowledge
- Develop the user interface
- Implement the complete system

Building an intelligent system: basic phases

Phase 6: Evaluation and revision of the system

- Evaluate the system against the performance criteria
- Revise the system as necessary

Building an intelligent system: basic phases

Phase 7: Integration and maintenance of the system

- Make arrangements for technology transfer
- Establish an effective maintenance program

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What we covered

- What intelligence and artificial intelligence mean.
- How AI developed from symbolic methods to modern generative AI.
- Different approaches to building intelligent systems.
- Major applications of AI in mining.
- The basic lifecycle of developing an intelligent system.

Key takeaways

- AI is not a single method, but a family of approaches to intelligent problem solving.
- Different problems require different AI approaches.
- The choice of method depends on the task, available data or knowledge, and system constraints.
- Building intelligent system is an iterative process: define the problem, select an approach, develop, evaluate, deploy and improve.

The next step is to understand how engineering problems are represented using data, search and computational problem-solving methods.

Thanks!